AI Image Classification System

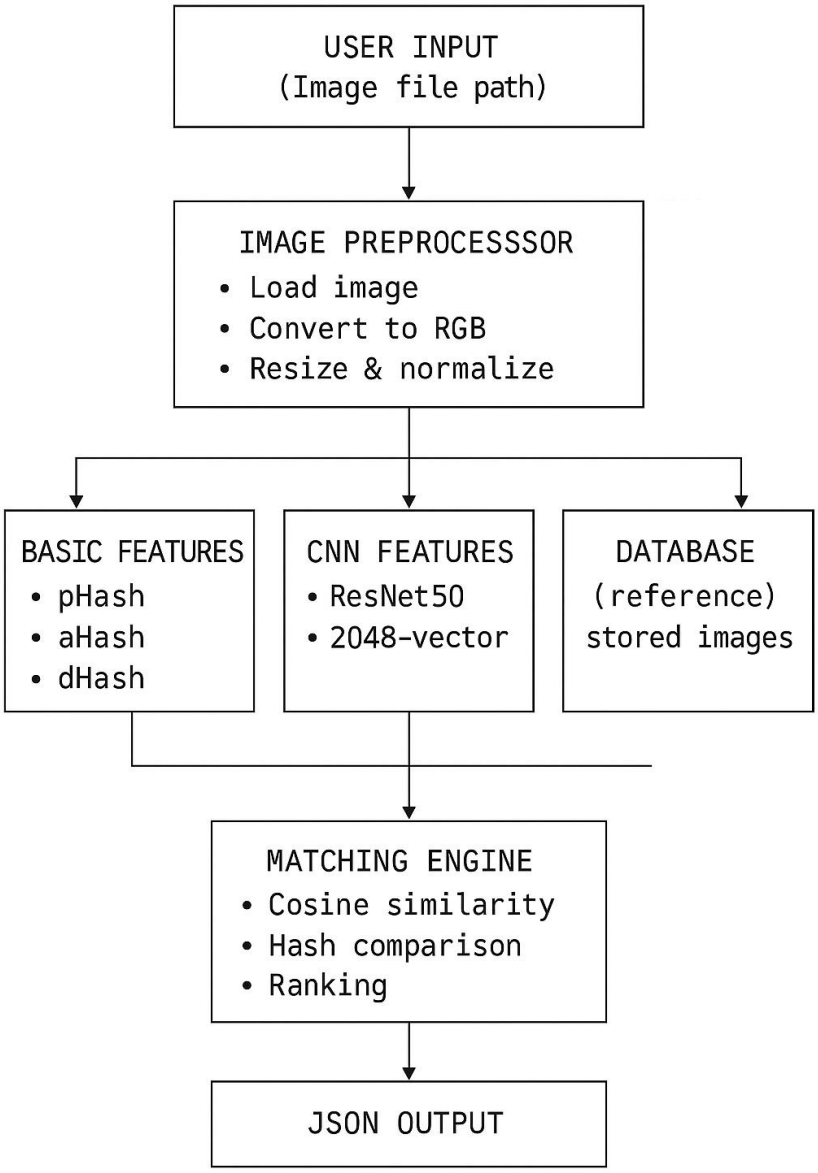
Project Overview

This project implements an **AI-based Image Classification System** capable of identifying the object in an input image and matching it with one of the predefined categories in a reference database.
It uses **two independent classification methods**:

1. **Basic Classifier** – hashing & color-based comparison
2. **AI Classifier** – deep learning feature extraction using **ResNet50**

Supported Categories

- Cat
- Dog
- Elephant
- Orange
- Watermelon
- Computer
- Rock


USER INPUT
(Image file path)
IMAGE PREPROCESSSOR
• Load image
• Convert to RGB
• Resize & normalize
BASIC FEATURES
• pHash
• aHash
• dHash
CNN FEATURES
• ResNet50
• 2048-vector
DATABASE
(reference)
stored images
MATCHING ENGINE
• Cosine similarity
• Hash comparison
• Ranking
JSON OUTPUT

Technologies Used

PROGRAMMING	
Main Language	Python 3.9+

LIBRARIES	
Image Loading	Pillow
Image Processing	OpenCV
Numerical Ops	NumPy
Hashing	ImageHash
Deep Learning	PyTorch
Pretrained Models	TorchVision

Method 1 — Basic Classifier

Uses traditional, non-AI feature extraction:

Extracted Features

- **pHash**
- **aHash**
- **dHash**
- **Color histogram**
- **Dominant colors via K-Means**

Method 2 — AI Classifier (ResNet50)

Image Preprocessing

- Resize to 256
- CenterCrop to 224
- Normalize using ImageNet mean/std

Internal Algorithms

Original Image
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Grayscale → 32×32
|

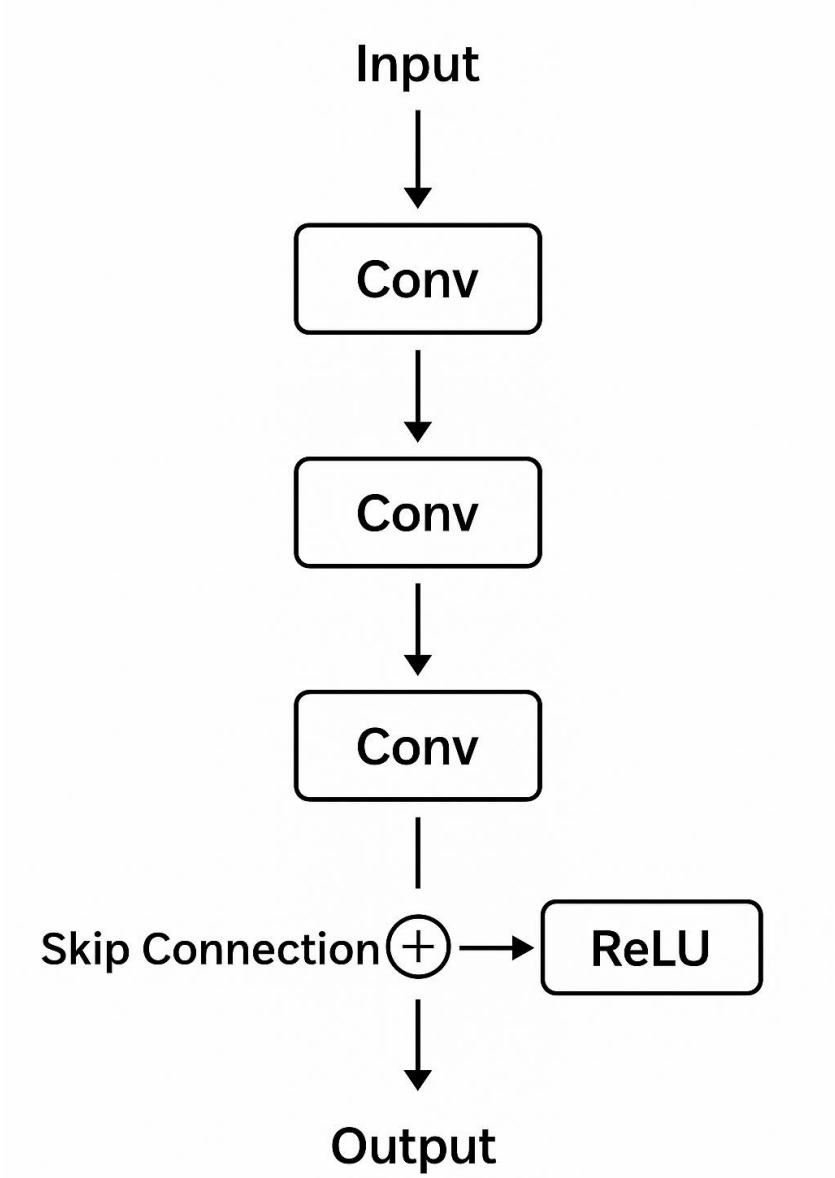
DCT
|

Take 8×8 low-frequency
|

Median threshold
|

64-bit binary hash

ResNet50 Residual Block



Accuracy Interpretation

Score	Meaning
85–100%	Very strong match
70–84%	High confidence

55–69%	Moderate match
<55%	Low certainty

Factors Affecting Performance

- Image quality
- Lighting
- Background noise
- Number of reference images

Minimum recommended dataset: **5–10 images per class**, optimal **15–25**.

Summary

This project demonstrates two complementary approaches:

Traditional Approach

- Fast
- Transparent
- Less accurate

Deep Learning Approach

- High accuracy
- Strong generalization

- Requires more computational power

Together they create a hybrid, reliable classification system suitable for both small-scale and advanced AI environments.